

Food Safety Management Systems

Science

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Food Safety Management Systems (A07221)

Short Title: Food Safety Management Systems
Faculty Science and Computing
Department Science
Cluster Food Science
Author EKENT
Credits 5

Level: Intermediate

Description of Module / Aims

This module describes the implementation of Food Safety Management Systems in the food processing industry. The importance of GHP and GMP as a pre-requisite to HACCP, risk analysis and food contamination are covered. The module prepares the student to have the knowledge and skills to implement a HACCP plan in the food industry, and to understand the role and importance of auditing in the food industry.

Programmes

QUAL -0026	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	3 / 6 / M
QUAL -0026	BSc in Agriculture (WD_SAGUL_D)	3 / 6 / M
QUAL -0026	BSc in Food Science with Business (WD_SFDSC_D)	3 / 6 / M

Indicative Content

- Food Safety Management Systems: HACCP, ISO22000, BRC etc.
- Food hazards: microbiological, chemical and physical
- Application of GHP and GMP as a pre-requisite to HACCP
- Examination of the key components of risk analysis, namely risk assessment, risk management and risk communication
- Origins of HACCP, basic principles of the HACCP system and its implementation in the food industry
- Auditing: typical audit sequence and roles

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise Food Safety Quality Management Systems.
2. Investigate microbiological, chemical and physical food hazards.
3. Appraise the importance of GHP and GMP as a pre-requisite to HACCP.
4. Apply the key components of risk analysis to the implementation of a HACCP plan.
5. Complete a HACCP Plan based on HACCP Principles and GMP.
6. Classify the various audit types, the audit process, corrective and preventative action.
7. Develop effective and efficient self-directed study skills.

Learning and Teaching Methods

- Lectures.
- Case studies.
- Site visit.
- Self-study of assigned reading.
- Guest speaker.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Assignment</i>	70%	2,3,4,5,7
<i>In-Class Assessment</i>	30%	1,6

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Field Trips	6	
Independent Learning	99	

Essential Material

- "Food Safety Authority of Ireland." www.fsai.ie
- "Module, Food Safety Management Systems." www.moodle.wit.ie
- Motarjemi, Y. and H. Lelieveld. *Food Safety Management: a practical guide for the food industry*. United States of America: Amsterdam: Academic Press, 2014.

Supplementary Material

- Adams, H.R. and M.O. Moss. *Food Microbiology*. Cambridge U.K.: RSC, 2008.
- Bolton, D. and J. Sheridan. *HACCP: for Irish Beef, Pork and Lamb slaughter*. Ireland: Teagasc, Agriculture and Food Development Authority, 2002.
- Motarjemi, Y. and H. Lelieveld. *Hygiene in Food Processing*. U.K.: Academic Press, 2014.